

Assignment - Module 8

Part A.

Define the following words and give an example in each

1. Probability - Probability is the likelihood of the occurrence given in numerical form (0-1). Example the probability of it raining in the month of March is 0.9 and the probability of it not raining is 0.1
2. Random Experiment - This is an experiment that can be repeated as often as possible as we wait and the outcome cannot be predicted example tossing a coin can be either H or T
3. Sample space - A sample of all logically possible outcomes resulting from an experiment example if a basket has oranges and mangoes and a fruit is picked randomly. If the fruits are equal, the sample space is $S = (\text{Orange, Mango})$
4. Event - This is a subset of a sample space of a random experiment.

Part B.

Application Question

Let A = High Shock ~~Scratch~~ Resistor
B = High Scratch Resistor

Shock \ Resistor	High	Low
Scratch Resistor - High	40	4
Scratch Resistor - Low	1	5

(a) Determine the number of samples $A \cap B$

Soln

Both are High $A \cap B = 40$

(b) Determine the number of samples in A'

Soln

$A' = \text{Not high Shock} = 9$

(c) Determine the number of samples in $A \cup B$.

Soln

$$A \cup B = 40 + 5 = 45$$

Part C

Empirical Probability

Blood Group distribution

Type	Frequency
A	22
B	5
AB	2
O	21
	<u>50</u>

using

$$P(A) = \frac{\text{Frequency of Event A}}{\text{Total Frequency}}$$

① Find probability of B.g type O = $\frac{21}{50}$

(ii) Type A or Type B

$$\frac{22}{50} + \frac{5}{50} = \frac{27}{50}$$

iii)

Neither A nor type B

$$\frac{2}{50} + \frac{21}{50} = \frac{23}{50}$$

(iv) Find probability that the person doesn't have type AB.

$$1 - P(AB) = 1 - \frac{2}{50} = \frac{48}{50} = \frac{24}{25}$$

Part D

Explain the following

i) $P(A) = -0.2$ $P(B) = 0.4$ $P(C) = 0.5$ $P(D) = 0.3$

since probability cannot be negative then this occurrence is not permissible

ii) $P(A) = 0.4$ $P(B) = 0.3$ $P(C) = 0.3$ $P(D) = 0.5$

summing up the total occurrence $\Rightarrow 0.4 + 0.3 + 0.3 + 0.5 = 1.5$ = Probability should be between 0 and 1 Hence not permissible for

Part E

Use addition rule to show

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Correcting the equation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Significance in real life applications

in Resource Allocation

Example the system is running tasks A and B

A : task requiring Machine 1

B : task requiring Machine 2

same tasks need both machines

the formula can be used to ensure accurate workload estimation and better solutions